From ML to LLM: Evaluating the Robustness of Phishing Web Page Detection Models against Adversarial Attacks

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Phishing attacks attempt to deceive users into stealing sensitive information, posing a significant cybersecurity threat. Advances in Machine Learning (ML) and Deep Learning (DL) have led to the development of numerous phishing web page detection solutions, but these models remain vulnerable to adversarial attacks. Evaluating their robustness against adversarial phishing web pages is essential. Existing tools contain datasets of pre-designed phishing web pages for a limited number of brands and lack diversity in phishing features.

To address these challenges, we develop `PhishOracle`, a tool that generates adversarial phishing web pages by embedding diverse phishing features into legitimate web pages. We evaluate the robustness of three existing task-specific models—Stack model, VisualPhishNet, and Phishpedia—against `PhishOracle`-generated adversarial phishing web pages and observe a significant drop in their detection rates. In contrast, a Multimodal Large Language Model (MLLM)-based phishing detector demonstrates stronger robustness against these adversarial attacks but still is prone to evasion. Our findings highlight the vulnerability of phishing detection models to adversarial attacks, emphasizing the need for more robust detection approaches. Furthermore, we conduct a user study to evaluate whether `PhishOracle`-generated adversarial phishing web pages can deceive users. The results show that many of these phishing web pages evade not only existing detection models but also users.

CCS Concepts: • **Security and privacy**; • **Computing methodologies** → **Artificial intelligence**; *Machine learning*;

Additional Key Words and Phrases: Phishing, Machine Learning, Deep Learning, Large Language Models, Adversarial Attacks

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1 Introduction

Phishing attacks extract sensitive information from targeted users. Attackers create phishing web pages and share URLs with users via social media platforms, e-mails, and so on. These requests deceive users into clicking on the URLs and redirect them to phishing web pages where they submit sensitive information. Attackers use this sensitive information to launch identity theft attacks or use information to exploit opportunities for financial

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profits. In Q1 2024, the APWG [1] recorded 963,994 phishing attacks, with the social media platforms being the most targeted sector, accounting for 37.4% of the attacks. Recent years have witnessed improvements in phishing web page detection approaches with the use of **Machine Learning (ML)** and **Deep Learning (DL)** methods.

The ML-based phishing web page detection solutions [2–5] consist of several stages. Initially, datasets containing both phishing and legitimate samples, including URLs, HTML content, and screenshots, are collected from publicly available repositories. Subsequently, relevant features are extracted based on the URL (such as URL length, presence of `@`, HTTPS, hyphens in the URL, etc.), web page content (such as internal and external hyperlinks, CSS styles, pop-up logins, `<form action=“”>` fields, etc.), and third-party features (such as PageRank, Google Index, etc.). Models such as Random Forest, Decision Tree, Support Vector Machine, and many others are trained on these datasets, with performance evaluated using metrics like accuracy, precision, recall, and F1-Score. Finally, the model's efficiency is validated on newly unseen phishing web pages.

With the continual progress in DL techniques, there is an enhanced capability to analyze web page screenshots and logos by employing state-of-the-art **Computer Vision (CV)** models [6–10]. DL techniques are utilized to identify brand logos on web page screenshots, comparing them with a reference brand list. This comparison considers similarity scores and domain analysis to classify the web page as phishing or legitimate.

Recently, **Large Language Models (LLMs)** have gained significant attention and adoption across various industries. These models pretrained on massive datasets are capable to understand and generate text, code, images, and videos. Therefore, the past 2 years have seen increasing application of LLMs to address different challenges in cybersecurity [11]. In the context of phishing, a recent work by Lee et al. [12] proposes an LLM-based pipeline that identifies the targeted brand by analyzing both HTML contents and web page screenshots. The results show that **Multimodal Large Language Models (MLLMs)** are helpful in outperforming a state-of-the-art solution, namely VisualPhishNet [8], which uses CV for similarity-based phishing detection.

Analyzing existing ML- and DL-based phishing web page detection approaches reveals significant challenges. Initially, researchers face time-consuming efforts for dataset collection containing phishing web pages with diverse phishing features. Furthermore, the short-lived nature of phishing web pages necessitates the extraction of both content-based and third-party-based features. Subsequently, it is essential to evaluate the robustness of these detection models to ensure that they can accurately classify new phishing web pages. Recent works [13–17] emphasize the importance of assessing the models against adversarial phishing web pages to determine their performance in detecting such sophisticated threats.

Existing phishing tools (BlackEye [18], ZPhisher [19], and ShellPhish [20]) contain pre-designed phishing web pages datasets that can be used to evaluate the robustness of phishing web page detection models. However, these datasets are limited to a small number of brands and lack diversity in phishing features. Moreover, these tools cannot generate multiple versions of phishing web pages with diverse sets of phishing features. We tested several brands such as Yahoo, Amazon, Instagram, Facebook, and Netflix from BlackEye [18]. However, since the tool provides repeated URLs with each execution, many of these URLs were already blocked, rendering the adversarial pages ineffective for assessing evasion attacks.

To overcome these limitations, we develop an automated tool—`PhishOracle`—that *generates* adversarial phishing web pages by embedding diverse phishing features into legitimate web pages. `PhishOracle` parses web page content and randomly selects a set of 12 content-based and 5 visual-based phishing features (refer Table 2), enabling the creation of diverse phishing web pages without being limited to a predefined set of brands. Due to this randomized process, `PhishOracle` can generate multiple phishing web pages for a single legitimate web page, each with a unique set of phishing features. Furthermore, `PhishOracle` is extensible, allowing the integration of new phishing features to adapt to evolving phishing threats. These generated adversarial phishing web pages are used to evaluate the robustness of existing phishing web page detection models.

In this article, we evaluate the robustness of phishing detection systems—that use conventional ML (Stack model [21]), DL (VisualPhishNet [8], Phishpedia [9]), and LLM-based phishing detector [12]—against `PhishOracle`-generated adversarial phishing web pages. Our results reveal that traditional phishing detection models experience

Table 1. Existing Phishing Tools vs. PhishOracle

Tools	Phishing Features															Static (S)/ Dynamic (D)
	action="local.php"	Save or Mail Credentials	Added/Removed <script> tags	Local CSS	Local JS	Modified Hypertext References	Font Style	Disable <a> tags	Disable other login <button>	Popup Login	Disabled Right Click	Disabled CTRL and Fn	Body Opacity	Replace blank space with character	Logo Transformations	
BlackEye [18]	✓	✓	✓				✓									S
ZPhisher [19]	✓	✓	✓	✓	✓											S
ShellPhish [20]	✓	✓	✓	✓	✓	✓	✓									S
PhishOracle	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	D

a significant drop in detection rate against the adversarial phishing web pages. While LLM-based phishing detector demonstrates higher resilience compared to these models, but it still experiences a decline in detecting adversarial phishing pages. We summarize our contributions as follows:

(1) We propose PhishOracle, a phishing web page *generator* capable of producing adversarial phishing web pages by randomly embedding content-based and visual-based phishing features into legitimate web pages. We carry out comprehensive evaluations to evaluate the robustness of ML-, DL-, and LLM-based phishing web page detectors using PhishOracle-generated pages. To the best of our knowledge, this is the first work to do so with one tool.
(2) We conduct a user study to evaluate the effectiveness of PhishOracle-generated phishing web pages generated to deceive users. The experiment results demonstrate that on average ~48% of these generated phishing web pages are incorrectly classified as legitimate by the users.
(3) We further validate the effectiveness of adversarial phishing web pages generated by PhishOracle by testing them against 90+ security vendors on VirusTotal.
(4) Finally, we contribute a dataset containing 9,067 legitimate web page screenshots, on which we manually labeled the identity logos. The PhishOracle code base, web app, generated phishing web pages, and survey screenshots are available on our GitHub repositories [22, 23].

Additionally, we develop PhishOracle web app, which allows users to selectively incorporate both content-based and visual-based phishing features into a legitimate web page. Compared to the existing phishing web page dataset tools, PhishOracle offers a broader range of features, including content-based manipulations such as pop-up logins and logo transformation techniques like adjusting opacity, adding watermarks, and more—capabilities that existing tools lack. Table 1 provides a comparative analysis of phishing web pages dataset tools and PhishOracle, highlighting the range of phishing features each tool supports. PhishOracle is a *gray-hat* tool, primarily designed to validate existing phishing web page detection solutions, which are inherently white-hat in nature. However, its capability to generate phishing web pages introduces a gray-hat aspect to its functionality.

The rest of this article is structured as follows: Section 2 reviews existing phishing web page dataset toolkits drawing a comparison with PhishOracle and discusses ML, DL, and LLM models for phishing web page detection. Section 3 defines the threat model, outlining the adversarial capabilities and objectives considered in our evaluation. Section 4 discusses PhishOracle, a tool that generates adversarial phishing web pages by adding randomly selected content-based and visual-based phishing features into legitimate web pages. Section 5 details the performance metrics and datasets used, including the creation of evasion dataset with PhishOracle. It also evaluates robustness of the Stack model, VisualPhishNet, Phishpedia, as well as the LLM-based phishing detector. Moreover, the section presents the evaluation of tools from different security vendors against these adversarial phishing web pages. Section 6 describes a user study assessing the effectiveness of PhishOracle-generated adversarial phishing web pages in deceiving users based on the visual appearance. It also outlines PhishOracle web app, which is designed to generate phishing web pages. Section 7 discusses the limitations of our approach. Finally, Section 8 concludes the article by discussing the performance of phishing web page detection models on PhishOracle-generated adversarial phishing web pages.

2 Related Work

In this section, we summarize the existing datasets of pre-designed phishing web pages, which are limited to a small set of brands. We contrast these with PhishOracle, which *generates* phishing web pages by embedding diverse phishing features into legitimate web pages. Additionally, we review existing ML-, DL-, and LLM-based phishing web page detection solutions.

2.1 Phishing Web Page Datasets

Numerous open source phishing tools such as BlackEye [18], ZPhisher [19], ShellPhish [20], and others offer pre-designed phishing web pages tailored to a small number of specific brands. For example, Shellphish [20] has a dataset containing pre-designed phishing web pages for 29 distinct brands. Similarly, ZPhisher [19] has pre-designed phishing web pages dataset for 33 brands with each web page incorporating external resources such as CSS, PHP, and JS files. Consequently, whenever there is a need for a phishing web page related to any of these brands, the same web page is consistently fetched from the dataset. These tools exhibit two limitations: (a) they cannot *generate* phishing web pages for brands not included in their predefined set, and (b) because they already have pre-designed phishing web pages for each brand, they cannot provide a phishing web page with diverse/new phishing features for the same brand. The tools are inherently *static*, confined to a fixed set of brands.

2.2 Phishing Web Page Detection Approaches

ML and DL techniques have significantly advanced phishing web page detection by training models on large datasets containing phishing and legitimate samples. These datasets include URLs, web page content, and screenshots. Phishing samples are obtained from repositories such as PhishTank[1] and OpenPhish,[2] and legitimate samples are collected from Common Crawl[3] and Stuff Gate.[4] Alexa[5] and Tranco [24] provide a list of top-ranked legitimate domains, the content of which can be scraped. Datasets like UCI[6] provide feature-extracted files containing both legitimate and phishing samples, including URLs, web page content, and third-party features. ML models use these features to identify patterns associated with phishing web pages, while DL techniques use neural networks to analyze more complex patterns, such as visual similarity in web page screenshots and logos.

[1]PhishTank, https://www.phishtank.com/index.php.
[2]OpenPhish, https://openphish.com/index.html.
[3]Common Crawl, https://commoncrawl.org/.
[4]Stuff Gate, https://stuffgate.com.websiteoutlook.com/.
[5]Alexa Top Websites, https://www.expireddomains.net/alexa-top-websites/.
[6]UCI, https://archive.ics.uci.edu/dataset/327/phishing+websites.

2.2.1 ML- and DL-Based Phishing Web Page Detection Approaches. Li et al. [21] proposed a Stack model (GBDT, XGBoost, and LightGBM) for phishing web page detection using URL- and HTML-based features. In this approach, Word2Vec model [25] is used to learn HTML strings and represent them as vector encodings. The Stack model outperforms phishing detection solutions [26–28] by achieving an accuracy of 96.45%. The performance of the Stack model is better than individual ML classifiers for phishing web page detection.

As DL techniques have evolved alongside the adoption of visual-similarity-based approaches in phishing web page detection, Fu et al. [6] proposed a visual similarity-based approach for phishing web page detection. The algorithm consists of three stages: (a) capturing a screenshot of the suspicious web page, (b) normalizing the screenshot to a fixed size, and (c) describing the normalized image to a signature consisting of the pixel color and coordinate features. The **Earth Movers' Distance (EMD)** algorithm is used to compute the visual similarity between the obtained visual signature and the legitimate web page signatures and classifies the suspicious web page as phishing or legitimate. However, this approach performs the classification at the pixel level and does not consider the text. As a result, it cannot detect visually dissimilar web pages.

Afroz and Greenstadt [7] proposed PhishZoo, a visual-based phishing web page detection approach based on the web page profile. A profile of a web page contains SSL, URL, HTML content and logo features. As a browser loads a web page, its profile is generated and compared with the stored profiles of legitimate web pages. If the loaded web page profile does not match with the legitimate SSL, URL, HTML content, and logos, then the user is warned of the web page possibly being phishing. PhishZoo has advantages over URL-based approaches but also has limitations, such as the approach failing to classify a web page containing previous logo versions, and if the logos are tilted over 30 degrees.

Similarity-based phishing detection approaches, such as VisualPhishNet [8], identify phishing attempts by analyzing the visual similarity between web page screenshots. VisualPhishNet utilizes a triplet **Convolutional Neural Network (CNN)** trained to embed images into a feature space, where visually similar web pages are mapped closer together. The model compares a given web page screenshot against a set of reference images from known legitimate brands. However, the approach has limitations: it may produce **False Positives (FPs)** (when different brands have visually similar web page layouts) and **False Negatives (FNs)** (when a phishing web page and its legitimate counterpart exhibit significant design variations despite targeting the same brand). Additionally, the reliance on visual similarity makes the system susceptible to adversarial attacks that modify key visual elements, while maintaining the phishing intent.

Lin et al. [9] propose Phishpedia to detect phishing web pages and identify its target brand in a reference brand list. Phishpedia adopts a two-step approach: *object detection* and *brand identification*. The object in this context refers to the *identity logo* present on a web page screenshot for a given brand. In the first step, the region proposal network [29] analyzes the screenshot layout and predicts boxes around logos on the screenshot. The Faster RCNN model, based on the Detectron2 framework (from Facebook [30]), selects the identity logo having the highest confidence score from the set of candidate logos. The next step is logo classification, wherein a ResNetV2 network [31] extracts diverse features from various logo variants belonging to the same brand. The classification task is trained on Logo2K+ dataset [32]. This trained ResNetV2 model is linked with a global average pooling layer to generate vectors that represent each logo. These vector representations are compared to logos in the reference brand list using cosine similarity [33]. The brand exhibiting the highest similarity score is identified as the target brand. Upon identifying the brand represented by a logo, the system verifies the legitimacy of the web page by cross-referencing its domain with the domain associated with the identified brand. Phishpedia outperforms EMD [6], PhishZoo [7] and LogoSENSE [34] in terms of brand identification, phishing detection, and runtime overhead.

Building on Phishpedia, Liu et al. [10] propose PhishIntention, which enhances phishing detection by incorporating *credential-taking* intention analysis alongside brand identification. Experiment results show that PhishIntention outperforms EMD [6], VisualPhishNet [8], PhishZoo [7], and Phishpedia [9] in phishing detection,

achieving higher precision while at similar recall. Its ability to extract and validate phishing intentions makes it a more effective solution against phishing attacks.

2.2.2 LLM-Based Approaches. With technological advancements, LLMs have gained attention in various sectors and their ability to extract contextual information from web pages has made them increasingly valuable in phishing detection. However, existing reference-based phishing detection solutions [9, 10] are constrained by their reliance on reference brand lists, limiting their scalability to a large number of brands. To address this, Li et al. [35] propose an LLM-assisted approach that extracts brand information from HTML content and determines whether a web page attempts to collect user credentials. By not relying solely on logos, this approach expands the phishing detection capability of existing reference-based solutions [9, 10]. In a recent work, Lee et al. [12] introduce a two-stage system (requiring no training) that utilizes MLLMs (Gemini, Claude3, and GPT-4) for phishing web page detection. In the first stage, the MLLM identifies the brand of the web page by analyzing various aspects of the web page (such as the HTML content and screenshot). In the second stage, an MLLM verifies whether the identified brand matches with the domain of the web page. This approach outperforms the existing screenshot-based phishing web page detection model VisualPhishNet [8].

MLLMs pretrained on massive datasets have the potential to assist or replace the existing models for phishing web page detection, as they do not require labeled data for training and continuous maintenance (retraining), while still having the capability to understand web page semantics. Therefore, we take a proactive approach and evaluate an LLM-based phishing detection approach [12] against evasion attacks.

2.2.3 Robustness of Phishing Detection Models against Adversarial Attacks. Recent works in phishing detection also concentrate on how the phishing detection models perform to classify adversarial phishing evasions. Sabir et al. [36] propose URLBUG, an adversarial URL generator designed to generate adversarial phishing URLs by targeting the domain, path, and top-level domain parts of URLs. These generated adversarial URLs are used to evaluate the robustness of several ML-based phishing URL detectors. The experimental results reveal that these adversarial URLs significantly challenge the ML-based phishing URL detectors, showing vulnerabilities and limitations.

Object detection models are also prone to adversarial attacks, allowing attackers to create adversarial logos that closely resemble the legitimate ones and evade logo-based phishing web page detectors [37]. Apruzzese et al. [14] examine the real-world impact of adversarial techniques on phishing web page detection models. Their findings reveal that attackers use simple, low-complexity methods such as blurring, cropping, and masking logos to evade ML-based phishing web page detectors. Another work by Lee et al. [15] generates adversarial logos by adding noise to brand logos in a reference brand list, enabling these logos to evade logo-based phishing web page detection solutions.

In another work, Apruzzese and Subrahmanian [13] introduce two sets of adversarial attacks: one where an attacker modifies only a few known features and another where the attacker has partial knowledge of the feature set. They develop a robust phishing web page detection model that withstands these attacks, demonstrating significant performance improvements over existing phishing web page detection models, which struggle against these adversarial attacks. Ji et al. [16] recently performed an extensive assessment of visual similarity-based phishing web page detection models. The robustness of these phishing detection models is also evaluated against adversarial phishing web pages that utilize visual modifications like rotation, repositioning, resizing, and blurring, showing that even slight modifications can degrade the detection accuracy of the models. To improve the model's robustness, the authors argue to combine text recognition with visual analysis and use preprocessing methods like scaling and denoising to reduce the effects of adversarial alterations.

Inspired by recent efforts to evaluate the robustness of the phishing web page detection models against adversarial phishing web pages [13–16], we develop a tool that generates such adversarial phishing web pages. This tool aims to enhance phishing web page detection models by identifying vulnerabilities through adversarial techniques, particularly targeting visual elements such as logos.

3 Threat Model

In this work, we consider a black-box adversary who aims to evade phishing web page detection models by generating adversarial phishing web pages using widely observed visual transformation techniques, particularly logo manipulations.

The adversary generates adversarial phishing web pages by applying logo-based transformation techniques, which include adjusting opacity, applying Gaussian blur, and overlaying mesh/watermarks. These transformation techniques have been widely used in prior works [14, 16, 38] to evaluate the robustness of visual-based phishing web page detection models.

The adversary's goal is to generate adversarial phishing web pages targeting brands, such that the generated phishing web pages evade detection. We assume that the adversary does not have access to deployed configurations or model weights. This setup aligns with a realistic black-box threat model. The adversary can use `PhishOracle` to modify the visual appearance of a legitimate web page and generate adversarial variants embedded with different visual transformations.

Thus, the attack strategy is to download a legitimate web page for a targeted brand, apply logo transformation techniques, generate adversarial phishing web pages, and evaluate the robustness of the phishing web page detection models. A successful evasion occurs when the model predicts the brand on the web page screenshot with low confidence, reducing the likelihood of detection.

Our target systems are phishing web page detection models that rely on visual-based features to identify the brand in a web page screenshot and compare it with the domain to detect phishing.

Out of Scope. We assume that attackers do not have access to deployed configurations or model parameters. As a result, advanced white-box attacks that require internal model access, such as gradient-based methods (e.g., FGSM [39], DeepFool [40]), are not considered in our work. Similarly, logo perturbation attacks like generative adversarial perturbations [15], although feasible in a black-box setting, are beyond the scope of our current work.

4 Proposed Phishing Web Page Generation Tool: `PhishOracle`

Attackers create phishing web pages to collect users' sensitive information. Typically, they modify specific HTML tags like `<form>`, `<a>`, `<link>`, `<script>`, and more, redirecting the acquired sensitive information to databases or e-mail addresses owned by the attackers. In addition to content-based web page features, maintaining style consistency is another critical aspect that can deceive users. Typically, all web pages in the same organization adhere to a uniform style. The overall style similarity focuses on the visual aesthetics of a web page, which can be characterized by various CSS design elements such as background colors, text alignments, font family, font size, and subtle variations in favicons and logos. When integrated into a legitimate web page, these visual-based features do not significantly alter its layout, but are sufficient to mislead users into believing that the phishing web page originates from the legitimate domain owner. The visual resemblance between the generated phishing web page and the legitimate styling is remarkably close, further deceiving users into perceiving the phishing web page as a replica of the legitimate one. However, for these phishing web pages to reach users—the actual targets of the attackers—they must first evade phishing web page detection models. To achieve this, adversarial phishing web pages are created with subtle modifications that allow them to evade phishing web page detection models, increasing the likelihood that they will reach the users and deceive them.

We propose `PhishOracle` to generate adversarial phishing web pages for any given legitimate web page by randomly embedding content-based and visual-based features (refer Table 2). Figure 1 depicts the workflow of `PhishOracle`, with two inputs: a legitimate URL and a set/count of phishing features to add to the input legitimate web page. First, the legitimate URL is requested to fetch and save the web page content and associated resources. Second, the algorithm parses the web page content to deduce the list of applicable phishing features

Table 2. List of Phishing Features Based on the Content and Visual Attributes in a Web page

Type	Naming	Phishing Features	Explanation
Content	*C1*	Hypertext reference	Updating `href` with: `href="#"`, "`#content`", "`#skip`", or "`Javascript:void(0)`"
	C2	Disable key functions	Restrict users to view source code by disabling *f11* and *Ctrl + U*
	C3	`href` lookalike characters	Alphabets in `href` value are replaced by lookalike characters E.g.: replacing `a` with a lookalike character `ā, á, ä`
	C4	Hide links appearing on status bar	This feature does not allow users to view the `href` link even on hovering on the link
	C5	Disable anchor tags	Does not allow a user to navigate to other pages by clicking on the anchor tags
	C6	Replace blank space with a character	Blank spaces present in the container elements of HTML like `h1, p, span` tags are replaced with characters with `style=color: transparent;`
	C7	Save credentials	The credentials entered in the HTML form input tags are stored in a local file
	C8	Disable other login buttons	Web pages containing login buttons like Google, GitHub, LinkedIn, etc., are disabled and the credentials entered in the visible login page are stored locally.
	C9	Pop-up Login	Clicking the login or sign-up button triggers a login page to appear. The action field in the `<form>` tag is then altered, to store the credentials locally
	C10	Pop-up Login by clicking on `a` tags	Clicking the anchor tags, opens up a login page and the credentials are stored locally
	C11	IFrame tag with login page	`<iframe>` tags are added with a login page, the credentials entered are stored locally
	C12	Add dummy tags	Dummy `<img>`, `<link>`, `<script>`, `<a>`, and `<div>` tags are added to increase the DOM structure of web page
Visual	*V1*	Body Opacity	CSS `opacity` adjusts web page transparency from 0 to 1
	V2	Text Styling	The `font-family` changes the style of web page text
	V3	Opacity on Logo	CSS `opacity` property makes the logo image transparent `<img>` tags containing images with `.png` and `.svg` extensions are considered and a `opacity` of 0.1 to 0.35 is added to make the logo transparent
	V4	Adding Watermark on logo	A watermark is added either on the bottom right corner or diagonally on logo image
	V5	Image Transformations	Techniques like adding (1) clockwise or anti-clockwise rotation, (2) Gaussian blur, (3) gray-mesh, and (4) noise, transforms the logo image

from Table 2. For example, if the web page content contains anchor tags (`<a>`), then relevant phishing features like *C1*, *C3*, *C4*, *C5*, and *C10* (refer Table 2) are embedded in the web page to generate its corresponding phishing web page.

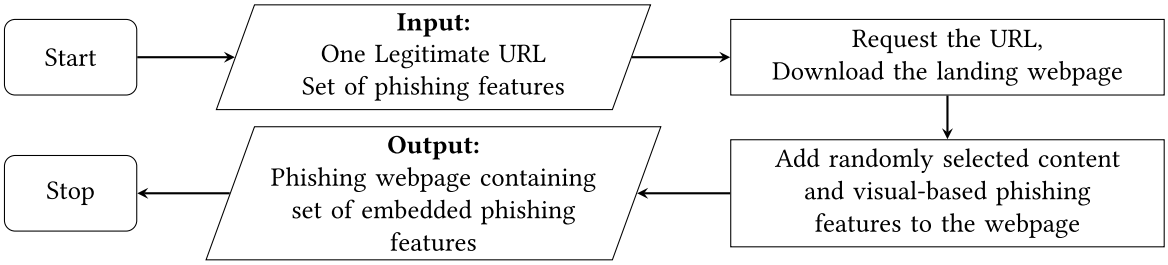
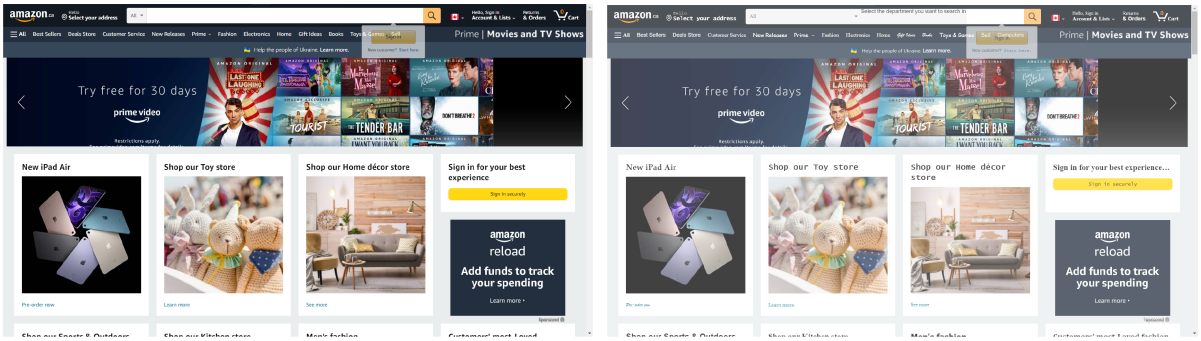


Fig. 1. Phishing web page generation tool.



(a) Legitimate webpage **(b)** Generated Phishing webpage

Fig. 2. PhishOracle-generated web page.

Figure 2 shows a legitimate web page and its corresponding phishing web page generated by PhishOracle by embedding content-based (*C2*, *C5*, *C7*, *C9*, and *C12*) and visual-based (*V1*, *V2*) phishing features, as listed in Table 2.

To produce a lookalike URL for the phishing web page generated by PhishOracle, a combination of *homoglyphs*, *prefixes*, and *suffixes* is utilized on the legitimate domain name. This combination results in the creation of domains that closely resemble legitimate domains, followed by corresponding URL parameters. For instance, if we consider the legitimate domain `facebook.com`, this process generates a lookalike domain such as `facebock-login.co`. A selection of homoglyphs examples includes substituting `d` with `cl`, `m` with `nn`, `w` with `vv`, `l` with `1`, `c` with `o`, `m` with `rn`, and so on. Additionally, there are prefixes like `secure-`, `logon-`, and `login-`, as well as suffixes such as `-login`, `-logon`, and `-secure`, and more to enhance the resemblance to legitimate domains.

5 Performance Evaluation

In this section, we outline the performance metrics used to evaluate phishing web page detection models. We detail the clean dataset used for model training and the evasion dataset—containing adversarial phishing web pages generated by PhishOracle—to evaluate the robustness of these models. Additionally, we discuss the evaluation of security tools against these adversarial phishing web pages.

5.1 Performance Metrics

The fundamental metrics to evaluate classifiers are **True Positive (TP)**, FP, **True Negative (TN)**, and FN. In the context of the phishing detection models, these metrics are used for determining the count of actual to predicted values, as shown in Table 3. The metrics *TP, FP, TN*, and *FN* are elaborated as follows:

Table 3. Confusion Matrix

		Predicted	
		Phishing	*Legitimate*
Actual	*Phishing*	True Positive	False Negative
	Legitimate	False Positive	True Negative

— *TP*: The model correctly identifies a phishing web site as phishing.
— *FP*: The model incorrectly identifies a legitimate web site as phishing.
— *TN*: The model correctly identifies a legitimate web site as legitimate.
— *FN*: The model incorrectly identifies a phishing web site as legitimate.

These metrics are used to compute the *Accuracy* ($\frac{TP+TN}{TP+FP+TN+FN}$), *Precision* ($\frac{TP}{TP+FP}$), *Recall* ($\frac{TP}{TP+FN}$), and *F1-Score* ($\frac{2\times Precsion\times Recall}{Precision+Recall}$) of classifiers.

5.2 Dataset Description

With technological advancements, attackers find new features to evade existing phishing web page detection solutions. Various solutions [21, 26–28] consider URL and HTML features, and a few solutions [6–9] use visual similarity features to classify new phishing web pages. With the current research focus on evaluating the robustness of phishing web page detection models against adversarial phishing web pages, we evaluate existing models for phishing web page detection using adversarial phishing web pages generated by `PhishOracle`.

In our study, we select the Stack model [21] due to its effectiveness in ML-based phishing web page detection to assess its capability to classify `PhishOracle`-generated adversarial phishing web pages that include a variety of phishing features. Next, we select VisualPhishNet [8], a screenshot-based phishing web page detection model that analyzes the visual similarity between web page screenshots to distinguish phishing web pages and legitimate ones by learning discriminative features based on brand identity and layout structure. Moreover, we select Phishpedia [9], known for its robustness in phishing web page detection and identifying target brands in web page screenshots. This makes it ideal for identifying brands in adversarial phishing web page screenshots generated by `PhishOracle`, incorporating different logo transformation techniques. Finally, we select an LLM-based phishing detector [12], selecting Gemini 1.5 Flash [41] as the LLM, to evaluate the solution's capability in identifying the target brand in HTML content and web page screenshots of `PhishOracle`-generated adversarial phishing web pages and verify whether the identified brand matches with the domain of the web page.

As the datasets to train these selected models [8, 9, 21] are outdated, we retrain them using our latest datasets (Table 4) and evaluate their robustness in detecting the adversarial phishing web pages generated by `PhishOracle`. We collect these dataset samples by using a Python script that includes various libraries, such as `Selenium`, `requests`, and `urllib.parse`. This script scraped 9,105 phishing samples from PhishTank[1], and 8,727 legitimate samples from Tranco [24], collected between December 2023 and January 2024. This collected dataset is referred to as $\mathcal{T}^1$ in Table 4. However, some brands in the reference brand list of the Phishpedia experiment prohibited downloading the web page content but allowed capturing screenshots, while others prohibited both. To ensure their inclusion in the dataset, we manually captured screenshots of these web sites. As a result, we collected a total of 9,067 legitimate web page screenshots, referred to as $\mathcal{T}^2$ in Table 4. Furthermore, we manually annotated the brand logos in each of the 9,067 web page screenshots to train the Faster RCNN model for brand identification in the Phishpedia experiment.

The collected samples in Table 4 are categorized into *CleanSet* and *EvasionSet*, each serving a distinct purpose for training and evaluating the robustness of the selected phishing web page detection models. Table 5 describes the categories of these *CleanSet* and *EvasionSet* datasets. *CleanSet*^{1} corresponds to the $\mathcal{T}^1$ dataset (refer Table 4),

Table 4. Dataset Collection Summary

Notation	Description	Collection Period	Sample Size	
			Phishing	Legitimate
$\mathcal{T}^1$	Contains URL, HTML content, and web page screenshots of phishing and legitimate samples	Dec 2023 to Jan 2024	9,105	8,727
$\mathcal{T}^2$	Contains URL, HTML content, and web page screenshot of legitimate samples, with manually annotated logos required to train the Faster RCNN model in the Phishpedia experiment	May 2024	0	9,067

which is used to train the Stack model [21] and to evaluate LLM-based phishing detector [12]. It includes 9,105 phishing samples and 8,727 legitimate samples, both containing URL, HTML content, and web page screenshots. *CleanSet*^{2} corresponds to the $\mathcal{T}^2$ dataset (refer Table 4), primarily used to train the Faster RCNN model for brand identification in the Phishpedia experiment [9]. It consists of 9,067 legitimate samples. Moreover, 909 phishing samples ($\subset \mathcal{T}^1$ dataset) and 82 legitimate samples ($\subset \mathcal{T}^2$ dataset) are specifically used to evaluate the performance of the brand identification and phishing detection of the entire Phishpedia system [9], as well as the LLM-based phishing detector [12]. *CleanSet*^{3} is used to train the VisualPhishNet model [8]. It consists of 896 phishing samples ($\subset \mathcal{T}^1$ dataset) covering 41 target brands, ~600 legitimate web page screenshots, and ~300 additional legitimate web page screenshots (from non-target brands $\subset \mathcal{T}^2$ dataset). This dataset is used to train the triplet CNN model for visual similarity-based phishing detection.

The *EvasionSet* datasets in Table 5—containing adversarial phishing web pages generated by `PhishOracle`—evaluate the trained models. *EvasionSet*^{1} contains 1,000 legitimate samples ($\subset \mathcal{T}^1$ dataset) with corresponding 1,000 adversarial phishing web pages generated using `PhishOracle` and is used to evaluate the Stack model [21], and the LLM-based phishing detector [12] (referred as *LLM-PD*^{H} in Table 6). *EvasionSet*^{2} consists of 82 legitimate samples ($\subset \mathcal{T}^2$ dataset), selected from Phishpedia's reference brand list, along with 170 adversarial phishing web pages generated using `PhishOracle`. This dataset is used to evaluate the Phishpedia system [9], as well as the LLM-based phishing detector [12] (referred as *LLM-PD*^{S} in Table 6). *EvasionSet*^{3} contains 107 phishing and 41 legitimate samples, which are a subset of *EvasionSet*^{2} and focus on the 41 target brands in VisualPhishNet [8]. This dataset is used to assess the robustness of the VisualPhishNet model against adversarial phishing web pages generated by `PhishOracle`.

5.3 Generating `PhishOracle` Validation Dataset

Developed using Python version 3.10.2, our `PhishOracle` tool uses various libraries, including `BeautifulSoup`, `requests`, `webbrowser`, `PIL`, and `urllib.parse`. For the Stack model experiment, we select 1,000 legitimate web pages ($\subset \mathcal{T}^1$ dataset) and embed relevant content-based and visual-based phishing features (refer Table 2) to generate corresponding 1,000 phishing web pages, forming the *EvasionSet*^{1} dataset. The algorithm includes a mechanism for detecting relevant tags within the web pages. The detection of such tags triggers the embedding of corresponding phishing features into the web page content. For instance, if the algorithm identifies `<a>` tag within the legitimate web page, it embeds features such as *C1*, *C3*, *C5*, *C10*, and *C12* (refer Table 2), resulting in the generation of the corresponding phishing web page. This entire process ensures the creation of balanced sets comprising legitimate and corresponding phishing web pages.

For VisualPhishNet, Phishpedia, and LLM-based phishing detection experiments, we select 82 legitimate web pages ($\subset \mathcal{T}^2$ dataset), all belonging to Phishpedia's reference brand list. We then generate 170 corresponding adversarial phishing web pages by incorporating visual-based features on logos (refer Table 2). Note that a single

Table 5. *CleanSet* and *EvasionSet* Description for Training and Evaluating the Phishing Web page Detection Models

Notation	Description	Sample Size		
		Phishing	Legitimate	PhishOracle (Generated Phishing)
$CleanSet^1$	$\mathcal{T}^1$ dataset used to train the Stack model [21] and to evaluate the LLM-based phishing detector [12]	9,105	8,727	0
$CleanSet^2$	$\mathcal{T}^2$ dataset used to train the Faster RCNN model, and ~900 phishing ($\subset \mathcal{T}^1$), and 82 legitimate samples ($\subset \mathcal{T}^2$), are used to evaluate the performance of brand identification and phishing detection model of the entire Phishpedia system [9], and the LLM-based phishing detector [12]	909	9,067	0
$CleanSet^3$	~900 phishing samples ($\subset \mathcal{T}^1$), and ~600 legitimate samples covering the 41 target brands + ~300 legitimate web page screenshots ($\subset \mathcal{T}^2$), from non-target brands, used to train the triplet CNN model in the VisualPhishNet model [8]	896	911	0
$EvasionSet^1$	Contains adversarial phishing web pages generated by PhishOracle for legitimate web pages selected from $\mathcal{T}^1$ dataset to evaluate the Stack model [21], and LLM-based phishing detector [12]	0	1,000	1,000
$EvasionSet^2$	Contains 170 adversarial phishing web pages generated by PhishOracle for 82 legitimate web pages ($\subset T^2$ dataset) in the reference brand list of Phishpedia [9] and to evaluate the performance of the Phishpedia [9] and the LLM-based phishing detector [12]	0	82	170
$EvasionSet^3$	Contains 41 legitimate and 107 corresponding adversarial phishing web pages ($\subset$ $EvasionSet^2$), and used to evaluate the performance of VisualPhishNet [8]	0	41	107

web page can have multiple PhishOracle-generated phishing web pages, each generated by embedding different visual-based features, specifically by incorporating logo transformation techniques (refer to visual-based features in Table 2). The $EvasionSet^2$ dataset comprises screenshots of these 82 legitimate and 170 generated phishing web pages, which are used to evaluate Phishpedia and LLM-based phishing detector. A subset of $EvasionSet^2$, referred to as $EvasionSet^3$, is constructed by selecting phishing and legitimate samples associated with the 41 target brands

Table 6. Performance of Stack Model, VisualPhishNet, Phishpedia, and *LLM-PD*^{H} and *LLM-PD*^{S}

Model	*CleanSet*				*EvasionSet*			
	Dataset	*Precision*	*Recall*	*F1-Score*	**Dataset**	*Precision*	*Recall*	*F1-Score*
Stack Model [21]	*CleanSet*^{1}	98.67%	98.32%	98.49%	*EvasionSet*^{1}	98.61%	70.86%	82.46%
VisualPhishNet [8]	*CleanSet*^{3}	98.42%	88.35%	93.11%	*EvasionSet*^{3}	98.73%	72.89%	83.87%
Phishpedia [9]	*CleanSet*^{2}	97.25%	81.63%	88.76%	*EvasionSet*^{2}	76.40%	40%	52.51%
LLM-PD^{H}	*CleanSet*^{1}	76.41%	73.83%	75.09%	*EvasionSet*^{1}	85.16%	67.55%	75.34%
LLM-PD^{S}	*CleanSet*^{2}	98.83%	97.25%	98.03%	*EvasionSet*^{2}	100%	79.51%	88.59%

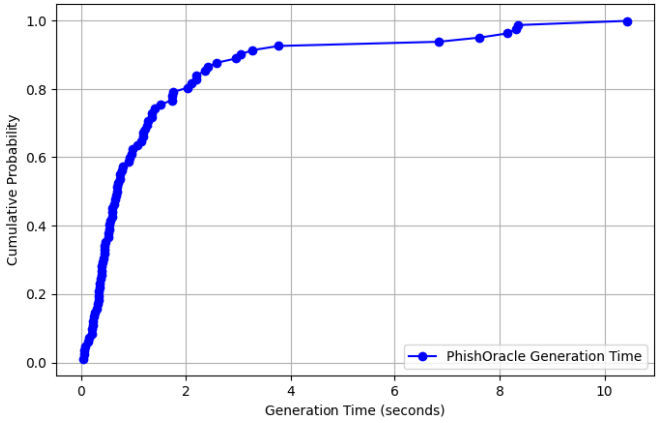


Fig. 3. CDF of PhishOracle processing time for generating adversarial phishing web pages in the *EvasionSet*^{2} dataset.

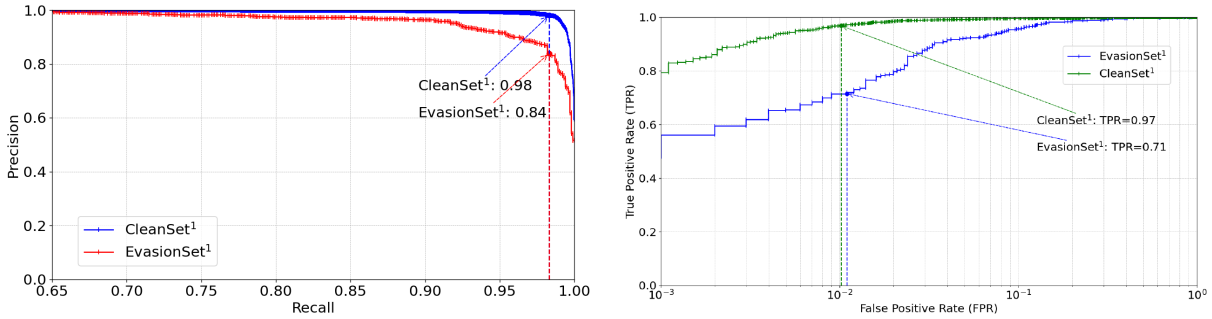
of VisualPhishNet. This dataset is specifically designed to assess the robustness of the VisualPhishNet model against phishing attempts targeting protected brands. Furthermore, we generate lookalike URLs for each of these generated phishing web pages (in *EvasionSet*^{1}, *EvasionSet*^{2}, and *EvasionSet*^{3}) using homoglyphs, suffixes, and prefixes. The source code and the datasets used in this experiment are available on our GitHub repository [22].

Figure 3 shows the performance evaluation of PhishOracle on the *EvasionSet*^{2} dataset. The average processing time taken to embed a diverse set of content and visual-based phishing features into legitimate web pages is ~1.48 seconds. The majority of the web pages are processed under 3 seconds, with only a few outliers extending up to 10 seconds due to potentially complex web page structures. The **Cumulative Distribution Function (CDF)** plot indicates that around 80% of web pages were processed in under 3 seconds, demonstrating the efficiency of PhishOracle to generate adversarial phishing web pages.

5.4 Validating Phishing Web Page Detection Models

In this section, we discuss the experiment setup for Stack model [21], VisualPhishNet [8], Phishpedia [9], and LLM-based phishing detector [12] and illustrate their performance on detecting adversarial phishing web pages generated by PhishOracle.

5.4.1 Stack Model. Li et al. [21] propose a Stack model for phishing web page detection using URL and HTML features, without considering third-party-based features (like DNS records, web traffic, etc.).



(a) Precision vs Recall **(b)** TPR vs FPR

Fig. 4. (a) Precision vs. recall and (b) TPR vs. FPR for the Stack Model.

Setup. To carry out our experiments, we use the feature extraction code of Stack Model made available by Phishpedia's authors on their GitHub repository [42], and we develop the Stack model using GBDT, XGBoost, and LightGBM classifiers.

Training and Testing the Model. We use *CleanSet*[1] containing URLs and HTML contents of 9,105 phishing and 8,727 legitimate web pages; and the train:test split is set to 80:20. In our experiments, Stack model achieves a high detection rate (recall) of ~98.32% at a high precision of ~98.67% on the*CleanSet*[1] dataset.

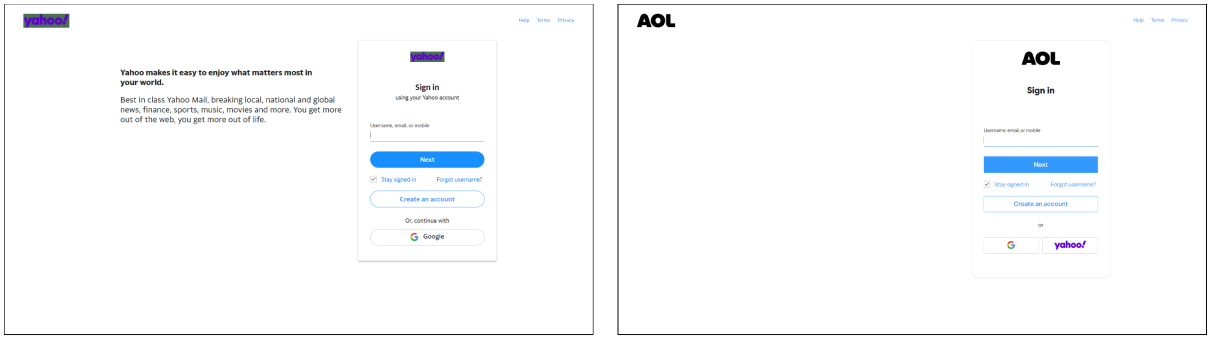
Validation on EvasionSet[1]. For evaluating the robustness of the Stack model, we now use the *EvasionSet*[1] that contains URLs and HTML content of legitimate and `PhishOracle`-generated phishing web pages. Stack model's detection rate falls to ~70.86% for a comparable precision of ~98.61%. Observe that, this is a significant reduction in comparison to the performance achieved on the test set *CleanSet*[1] (as mentioned above)—~28% in recall for about the same precision. Therefore, the Stack model is not robust to simple feature modifications (refer Section 4) and confirms the observations made in a recent study [43].

Figure 4(a) displays the precision–recall curve, illustrating the detection efficacy of the Stack model on the *CleanSet*[1] and *EvasionSet*[1] datasets. For a recall of ~98%, the Stack model's precision drops significantly on the *EvasionSet*[1] to ~84% in contrast to ~98% on the *CleanSet*[1]. This performance reduction indicates the model's reduced effectiveness in detecting adversarial phishing web pages generated by `PhishOracle`. In addition, Figure 4(b) displays the ROC curve for both datasets, drawn on a logarithmic scale to show the model's performance at low **False-Positive Rates (FPR)**. For a fixed FPR of $\sim 10^{-2}$, the model achieves a **True-Positive Rate (TPR)** of ~97% on the *CleanSet*[1], which drops to ~71% on the *EvasionSet*[1], indicating reductions in both precision and TPR for adversarial phishing web pages.

5.4.2 VisualPhishNet. Abdelnabi et al. [8] propose VisualPhishNet to detect phishing web pages by analyzing the visual similarity between web page screenshots using a triplet CNN model.

Setup. We initiate the experiment by cloning the baseline model from the Phishpedia GitHub repository [42].

Training and Testing the Model. In this experiment, we use phishing and legitimate web page screenshot datasets. The *CleanSet*[3] dataset consists of ~900 phishing samples ($\subset \mathcal{T}^1$ dataset) targeting 41 brands, along with ~600 legitimate web page screenshots for these 41 target brands, collected by visiting their official web site hyperlinks. Additionally, to ensure an unbiased training process, we include ~300 legitimate web page screenshots from non-target brands ($\subset \mathcal{T}^2$ dataset). This dataset is used to train the triplet CNN model in VisualPhishNet for visual similarity-based phishing detection. We follow the 60:40 train–test split, as suggested in [8], to evaluate the model's performance effectively. The retrained model achieves a precision of ~98.42% at a recall of ~88.35% on the *CleanSet*[3].



(a) Adversarial Phishing Webpage targeting Yahoo **(b)** Legitimate Webpage of AOL

Fig. 5. VisualPhishNet misclassify adversarial phishing web page targeting (a) Yahoo as (b) AOL due to high similarity in their visual appearance.

Validation on EvasionSet^3. Finally, we evaluate the trained VisualPhishNet model on the *EvasionSet^3*, containing 41 legitimate and 107 adversarial phishing web pages generated by `PhishOracle`. This results in a considerably lower phishing detection rate (recall) of ~72.89% at a precision of ~98.73%. This demonstrates that the adversarial phishing web pages generated by `PhishOracle`, which incorporate logo transformation techniques (such as watermark, rotation, opacity, etc.), evade the VisualPhishNet model. For example, as shown in Figure 5, the model incorrectly identifies an adversarial phishing web page targeting Yahoo (Figure 5(a)) as **America Online (AOL)** (Figure 5(b)), due to similarities in the visual appearance of the web page layouts, including logo placement and form fields.

Observations. To evaluate the robustness of VisualPhishNet, we use *EvasionSet^3*, containing adversarial phishing web pages generated by `PhishOracle`. The detection rate of VisualPhishNet drops to ~72.89% for a comparable high precision of ~98.73% (refer Table 6). Observe that this is a significant reduction compared to its performance on the test set *CleanSet^3* (as mentioned above)—~16% decrease in recall for about the same precision. This indicates that VisualPhishNet is affected by the logo-based adversarial manipulations. These findings align with recent evaluations [16], showing that the screenshot-based models like VisualPhishNet struggle with accurate brand identification when logos are altered, removed, or replaced.

5.4.3 Phishpedia. Lin et al. [9] propose Phishpedia to detect phishing web pages targeting a specific brand in a reference brand list.

Setup. We initiate the experiment by cloning the Phishpedia GitHub repository [42]. To carry out this experiment on Phishpedia, we manually labeled identity logos on 9,067 legitimate web page screenshots in the $\mathcal{T}^2$ dataset.

Training and Testing the Model. In this experiment, we consider two datasets. For evaluating the object detection task, we use $\mathcal{T}^2$ dataset (described in Table 4), containing 9,067 legitimate URLs and web page screenshots, and split it into 7,539 samples for training and 1,528 for testing. This results in a train:test split of 83:17. For the brand identification task, which uses the Siamese model pretrained on the Logo2K+ dataset [32], and the phishing detection task, we use *CleanSet^2*, containing 909 phishing web page screenshots with manually labeled identity logos and 82 legitimate web page screenshots from the reference brand list, with a train:test split of 80:20.

Results. The logo detection task achieves the mean Average Precision [44] of ~43.42%, which is considered good in object detection. The brand identification model accurately detects the brand on ~97.77% of phishing web pages. In phishing detection, the entire Phishpedia system achieves a precision of ~97.25% at a recall of ~81.63%. The values are slightly lower than the precision of ~98.2% at a recall of ~87.1% reported in the

paper [9]; the reasons could be that, we have a smaller dataset for training in our work, or there is a difference in the quality of data. However, close to ~81.63% recall at a high precision of ~98% is still a good performance.

Validation on EvasionSet^2. Finally, we evaluate the newly trained Phishpedia system on the *EvasionSet^2*, containing 82 legitimate web page screenshots and 170 `PhishOracle`-generated adversarial phishing web page screenshots. This results in a precision of ~76.40% at a recall of ~40% for phishing detection (refer Table 6). Observe that the `PhishOracle`-generated adversarial phishing web page screenshots containing logo transformation techniques (such as adding watermark, noise, etc.) can evade brand identification in the Phishpedia model.

5.4.4 Multimodal LLMs. With the emergence of LLMs, we today have several MLLMs that can process, analyze, and generate texts, images, and other multimedia contents. The capabilities of the MLLMs to analyze texts and images make them suitable for our task of brand identification in HTML content and web page screenshots. In a recent work, Lee et al. [12] propose a two-stage system of MLLMs for phishing web page detection. In the first stage, an MLLM analyzes various aspects of web page, including its HTML content and screenshots, to identify the brand. In the second stage, an MLLM verifies whether the identified brand matches the domain of the web page (URL), flagging mismatches as phishing. The approach employs separate prompts for analyzing HTML content and web page screenshots to identify the brand, along with an additional prompt to verify whether the identified brand matches the domain of the web page. In our study, we adopt these prompts to evaluate the LLM-based phishing detection pipeline [12] using Gemini 1.5 Flash [41] on both clean and adversarial phishing web pages, using *CleanSet^1* and *CleanSet^2*; and *EvasionSet^1* and *EvasionSet^2*, respectively.

To ensure a fair comparison with existing phishing detection models, we utilize two Gemini-based variants: *LLM-PD^H* and *LLM-PD^S* from [12]. The Stack model, which relies on URL and HTML-based features, is evaluated on *EvasionSet^1*. Accordingly, we use the HTML prompt from [12] and provide the corresponding HTML contents as input to the LLM-based pipeline, referring to this variant as *LLM-PD^H*. Similarly, Phishpedia, which identifies brands in web page screenshots is evaluated on *EvasionSet^3*—so we use the screenshot prompt from [12] and input the corresponding web page screenshots into the LLM-based pipeline, referring to this as *LLM-PD^S*. This approach ensures that the performance of LLM-based approach is directly compared to state-of-the-art phishing web page detection models on the same evasion datasets.

(1) Brand identification from *HTML content*

Setup. This experiment evaluates the ability of the LLM-based pipeline from [12] using Gemini [41] to identify brands from the HTML content of web pages. We employ the HTML-based prompt from [12] and refer to this variant as *LLM-PD^H* (as shown in Table 6). The model is tested on the *CleanSet^1* dataset, which comprises phishing and legitimate samples. We evaluate the performance of *LLM-PD^H* in brand identification and verifying its consistency with the domain of the web page. The model achieves a precision of ~76.41% at a recall of ~73.83%, and an F1-Score of ~75.09%.

Robustness Evaluation on EvasionSet^1. To assess the robustness of *LLM-PD^H* against adversarial phishing web pages, we evaluate its performance on *EvasionSet^1*, which contains phishing samples generated by `PhishOracle`.

Results. On the *EvasionSet^1* dataset, *LLM-PD^H* achieves a precision of ~85.16% at a recall of ~67.55%, and an F1-Score of ~75.34%.

Observations. LLM-PD^H achieves a precision of ~85.16% at a recall of ~67.55%, and an F1-Score of ~75.34% on *EvasionSet^1*, indicating a decline in recall compared to the *CleanSet^1*. This indicates an increase in FNs, where more phishing web pages are misclassified as legitimate. However, this corresponding increase in precision offsets this drop, resulting in a comparable F1-Score. These results suggest that the LLM-based phishing detection pipeline remains relatively robust against adversarial phishing attacks, although further improvements may be necessary to enhance recall.

(2) Brand identification from *web page screenshots*

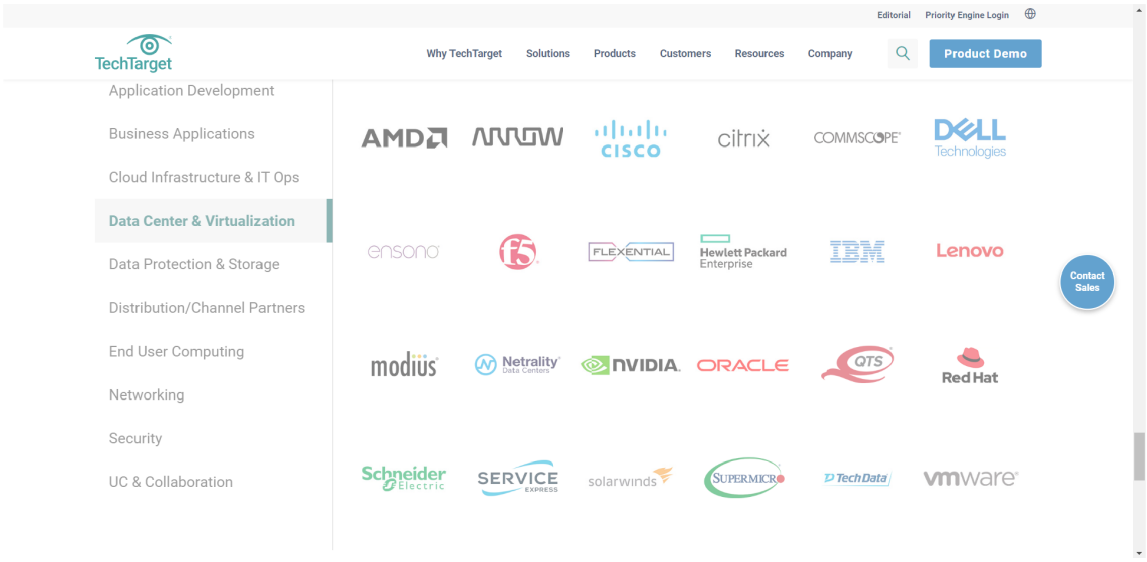


Fig. 6. Web page screenshot containing multiple logos of different brands.

Setup. In this experiment, we evaluate the ability of the LLM-based phishing pipeline [12] to identify brands from web page screenshots. We use the screenshot-based prompt from [12] and refer to this variant of Gemini as *LLM-PD*^{S} (as shown in Table 6). The model is tested on the *CleanSet*^{2}, a dataset consisting of both legitimate and phishing web page screenshots. We analyze how effectively *LLM-PD*^{S} identifies brand names and verifies them against the domain of the web page. The model achieves an impressive precision of ~98.83% at a recall of ~97.25%, and an F1-Score of ~98.03%.

Robustness Evaluation on EvasionSet^2. To assess the robustness of *LLM-PD*^{S}, we evaluate its performance on *EvasionSet*^{2}, which contains screenshots of legitimate web pages along with their corresponding adversarial phishing web pages variants generated by `PhishOracle`.

Results. Evaluation of *LLM-PD*^{S} on *EvasionSet*^{2} results in a precision of 100% at a recall of ~79.51%, with an F1-Score of ~88.59%.

Observations. *LLM-PD*^{S} is robust on the adversarial phishing web pages generated by `PhishOracle`, but it experiences a drop of ~10% F1-Score on *EvasionSet*^{2} as compared to *CleanSet*^{2}. For instance, in an adversarial phishing web page targeting Yahoo (Figure 5(a)), the first-stage MLLM correctly identifies the brand and provides supporting evidence. The second-stage MLLM while comparing the identified brand with the domain name/URL (http://signup-yahoo.com) incorrectly associates it with the identified brand as belonging to the same company, leading to evasion. Nevertheless, *LLM-PD*^{S} outperforms VisualPhishNet [8] and Phishpedia [9] in effectively identifying brands on adversarial phishing web pages generated by `PhishOracle`. Furthermore, *LLM-PD*^{S} efficiently identify the actual brand name in web page screenshots containing logos of multiple brands. As an illustration, it correctly identifies the brand name as "TechTarget" in the web page screenshot containing logos of multiple brands, as shown in Figure 6. Once the brand is correctly identified, comparing the URL's domain name with the identified brand's domain name gives the detection result.

Table 6 provides a comprehensive comparison of the Stack model, VisualPhishNet, Phishpedia, and *LLM-PD*^{H}, and *LLM-PD*^{S} in phishing web page detection.

Takeaways.

(i) The detection rate of the Stack model [21] drops by around 28% on PhishOracle-generated *EvasionSet*[1] compared to its performance on *CleanSet*1 , when evaluated at a comparable precision.
(ii) VisualPhishNet [8] experiences a drop of around 16% in detection rate when evaluated on *EvasionSet*[3] relative to its performance on *CleanSet*[3], when tested on a comparable precision.
(iii) Phishpedia [9] suffers a drastic drop of approximately 40% in detection rate on *EvasionSet*[2] due to adversarial logo transformations (refer to Table 2), at a 20% lower precision. This indicates that transformed logos can evade Phishpedia. Another contributing factor could be the smaller training dataset used in our work.
(iv) Compared to other phishing detectors, LLM-based phishing detector (*LLM-PD*[S]) (with no task-specific training) is more robust against adversarial phishing web pages generated by PhishOracle. Nonetheless, it also experiences a decline of performance—around 10% in F1-Score on *EvasionSet*[2] compared to *CleanSet*[2].

5.5 Evaluation of Security Tools against Adversarial Phishing Attacks

In this section, we present a 9-day evaluation of adversarial phishing web pages generated by PhishOracle, focusing on their detection by security tools.

Setup. We conducted a 9-day study to evaluate effectiveness of the security tools in detecting the adversarial phishing web pages generated by PhishOracle. The evaluation focused on VirusTotal [45], with more than 90 security vendors, along with GSB and Microsoft Defender SmartScreen. We initiate the experiment by generating one adversarial phishing web page for eight brands, including Microsoft, Yahoo, Store Steam, TalkTalk Group, UIBK, UChile, Trane, and LibreTexts. These web pages are designed by embedding diverse content-based features—such as altering form action fields, embedding pop-up login forms, using Javascript-based navigation prevention to restrict user movement—and visual-based features, including opacity and logo transformation techniques (refer Table 2). Once generated, these adversarial phishing web pages are hosted on GitHub Pages, allowing them to be publicly accessible for evaluation.

Evaluation Methodology. The adversarial phishing web pages generated by PhishOracle were examined over a 9-day span at 24-hour intervals. The total number of security vendors in VirusTotal flagging these web pages as phishing or suspicious were documented. Additionally, the identification by GSB and Microsoft Defender SmartScreen was also tracked to observe how promptly these popular security solutions recognize these adversarial phishing web pages.

Results. With a total of 96 security vendors on VirusTotal, the detection rates varied significantly among different brands. On Day 1, only one phishing web page targeting LibreTexts was flagged by only one security vendor (Trustwave). The other phishing web sites were still not marked by any of the security vendors. On Day 2, the Microsoft phishing web page was flagged by 6 security vendors on VirusTotal, and the Microsoft Defender SmartScreen marked it as "Dangerous Site." Several other phishing web pages targeting brands such as TalkTalk Group, UIBK, UChile, Trane, Steam, and Yahoo were flagged by 13 vendors, such as GSB, Trustwave, Webroot, and CyRadar. The phishing web page from LibreTexts was marked as "Dangerous" by GSB. On Day 3, the number of detections rose for phishing web pages targeting popular brands such as Microsoft and Yahoo, which were flagged by 18 and 16 security vendors, respectively. Moreover, the LibreTexts phishing web page was marked as "Dangerous" by Microsoft Defender SmartScreen; however, it remained unflagged by most of the security vendors on VirusTotal. From Days 4–9, the count of vendors marking these hosted adversarial phishing web pages remained stable. Figure 7 displays the detection rates of security vendors against adversarial phishing web pages generated by PhishOracle.

Observations. The 9-day evaluation study showed that phishing web pages targeting well-known brands (e.g., Microsoft, Yahoo) were flagged by more security vendors than those targeting lesser-known brands (e.g., LibreTexts). However, detections were alarmingly slow, with the majority of security vendors failing to identify phishing web pages within the first 24 hours. For instance, the Microsoft phishing web page was flagged by only

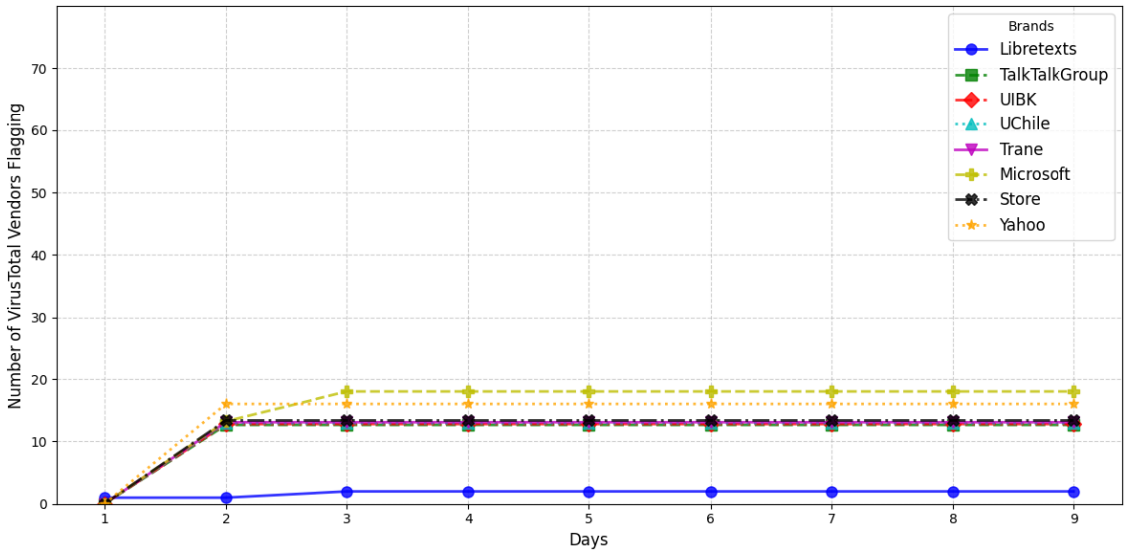


Fig. 7. Evaluation of security tools over 9 days against adversarial phishing attacks.

six vendors by Day 2, meaning over 90% of the 96 VirusTotal security vendors failed to detect it within the first day. Similarly, the LibreTexts phishing web page remained largely undetected, with just one vendor identifying it by Day 1.

These results align with prior research [46], which reported an average phishing site lifespan of 2.25 days, showing that phishing web pages remain active long enough to deceive victims. Detection rates in our study followed a similar trend, where most phishing web pages were flagged only after 2 or more days, and detections stagnated after Day 3. This delay is particularly concerning because even a few hours of undetected activity provides ample time for attackers to successfully execute phishing campaigns.

Our findings further support previous work [47], which highlights inconsistencies in how VirusTotal vendors classify phishing URLs. While some vendors flagged phishing web pages within 2 days, most security solutions failed to detect them within the first 24 hours, exposing critical detection gaps. Even for high-profile brands like Microsoft and Yahoo, only 16–18 out of 96 security vendors flagged them by Day 3, meaning ~80% of security vendors still failed to detect them by this time. This detection gap left phishing web pages accessible to potential victims for an extended period.

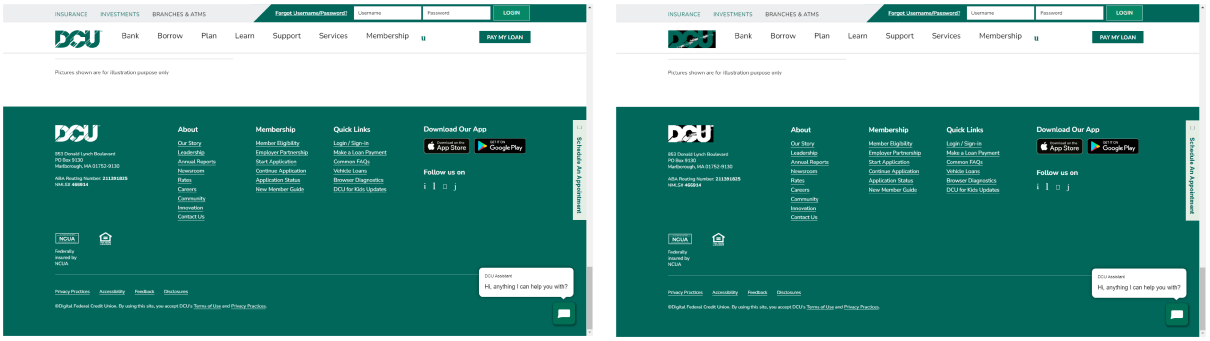
6 User Study and PhishOracle Web App

In this section, we discuss a user study to verify whether the phishing web pages generated by PhishOracle truly deceive the actual users. Moreover, we describe our PhishOracle web app, which is designed for both user education and phishing research purposes.

6.1 User Study

In this section, we assess the effectiveness of adversarial phishing web pages generated by PhishOracle in deceiving users. We conduct the study to evaluate the ability of users to determine the legitimacy of a web page based solely on its visual appearance, without relying on the URL.

Participants. The user study aims to determine whether PhishOracle-generated adversarial phishing web pages can deceive users based on their visual appearance alone, even if detection models can be evaded. To



(a) Legitimate webpage **(b)** Generated Phishing webpage

Fig. 8. User study: (a) Legitimate and (b) PhishOracle-generated web page screenshots.

achieve this, we selected 52 participants within the age range of 19 and 26, consisting of academic students and IT professionals. A recent study [48] examined whether phishing web pages that evade the phishing web page detection models can still deceive users based on visual appearance alone, which closely aligns with our research objective. Furthermore, the study suggests conducting a brief user study by presenting a limited number of phishing web page samples to the participants for evaluation. This selection allowed us to assess whether IT professionals—who are generally more knowledgeable about phishing threats and frequently encounter suspicious e-mails—can analyze a web page's visual appearance to determine its legitimacy. For academic students, the study also served as a form of user education, emphasizing the importance of inspecting web page visuals for phishing indicators. We adhered to the ethical guidelines [49] to safeguard user information confidentiality.

Research Question. The core research question (RQ) of the study is: *Can a user classify a web page as "phishing" or "legitimate" when presented with a screenshot with its address bar masked?* To answer this, participants were shown web page screenshots and asked to classify them as either phishing or legitimate. The goal is to assess whether users can distinguish phishing from legitimate web pages based solely on visual elements, without relying on the URL. While we acknowledge that real-world users typically have access to the address bar, prior report [50] has shown that many do not actively use URLs to verify a web site's legitimacy. This report found that 25.5% of users clicked on phishing links, and 18% went further to submit their credentials, indicating that a significant portion of users do not carefully inspect URLs before interacting with web pages.

Setup. We initiate the experiment by selecting 100 legitimate web pages and generating 100 corresponding adversarial phishing web pages using PhishOracle, by embedding randomly selected visual-based features (refer Table 2). Figure 8(a) and (b) showcases exemplar screenshots of legitimate and PhishOracle-generated phishing web pages, respectively, employed in the user study. The phishing web page screenshot (Figure 8(b)) is generated by embedding *V4* visual-based phishing feature into the corresponding legitimate web page (Figure 8(a)).

Distribution of Samples. To facilitate this, we utilize Google Forms, offering two options for each web page screenshot: "Phishing" or "Legitimate." We create 11 Google Forms by randomizing the arrangement of legitimate and PhishOracle-generated adversarial phishing web page screenshots. Employing Google Forms, we administer the user study to the 52 participants. The first nine forms comprise a mix of 10 legitimate and 15 generated phishing web page screenshots. One form featured 10 legitimate and 16 generated phishing web page screenshots, while another included 10 legitimate and 20 generated phishing web page screenshots. The URL field in each web page screenshot is masked to ensure unbiased classification outcomes.

Results. Evaluating the feedback of participants reveals that, on average, ~48% of the phishing web pages generated by PhishOracle are misclassified as legitimate. The user study results are presented in Figure 9.

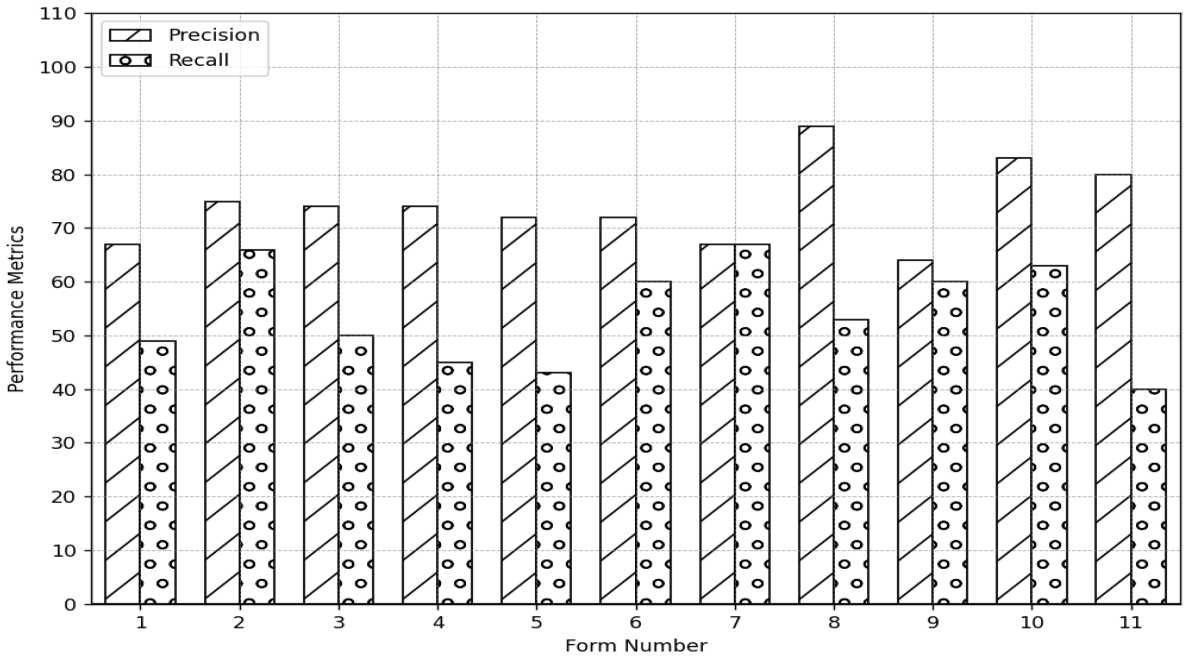


Fig. 9. Results of our user study.

Observations. Our findings from the experiments (refer Section 5.4) and the user study emphasize that the logo transformation techniques (such as opacity, rotation, blur, noise, gray-scale mesh) not only affect the performance of the phishing web page detection models but also deceive the users. This aligns with the work by Yuan et al. [51], who argue that assessing the user's perception of adversarial phishing web pages is a necessary step in phishing web page detection, as users are the actual targets.

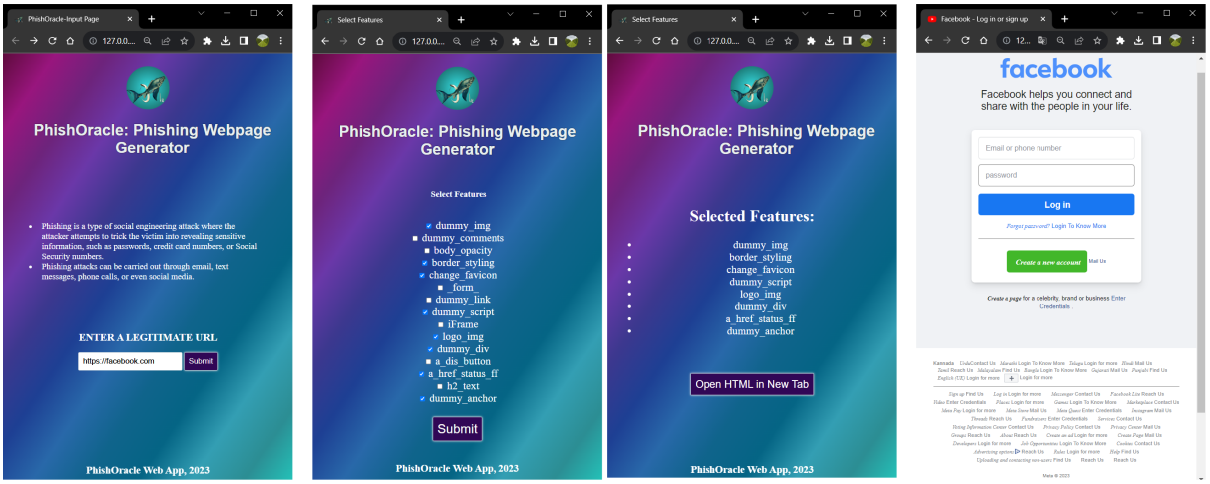
Limitations. In our user study, the web page screenshots do not include URLs, which could assist users in assessing the overall legitimacy of the web page. Instead, users are presented with web page screenshots and tasked to classify the web page as "phishing" or "legitimate" based solely on visual appearance.

6.2 `PhishOracle` Web App

We next introduce our `PhishOracle` web app, an interactive tool developed to serve both user education and phishing research purposes. This web app facilitates the process of generating phishing web pages by utilizing a legitimate input URL.

The web app was initially hosted on PythonAnywhere,[7] a cloud-based platform designed for hosting Python web apps, ensuring accessibility and reliability. However, the hosted web app was disabled due to a violation of the Terms and Conditions of PythonAnywhere as the content is related to phishing. We then hosted the web app on GitHub pages,[8] but the generated phishing web page was marked "Dangerous Site" because of the application containing phishing activities. The web app is thus added as a repository on GitHub [23], which can be downloaded and hosted on a local system. The web app comprises three rendering HTML templates: `index.Html`, `select_features.Html`, and `generate_webpage.Html`. The `index.Html` (Figure 10(a)) serves as the user's entry point, offering an input text box for legitimate URL submission. Once a URL is provided, the

[7]PythonAnywhere, https://www.pythonanywhere.com/.
[8]GitHub Pages, https://pages.github.com/.



(a) Index Page **(b)** Feature Selection Page **(c)** Generate Page **(d)** Phishing Webpage

Fig. 10. Generating phishing web page via `PhishOracle` web app.

application fetches the source code of the associated web page, extracts relevant features, and presents them as checkboxes on the `select_features.Html` page (Figure 10(b)), allowing users to select features to incorporate into the web page. After selecting features, the application embeds the chosen features into the legitimate web page to generate the corresponding phishing web page, which can be viewed on the `generate_webpage.Html` page (Figure 10(c)). Users can open the generated phishing web page (Figure 10(d)) in a new browser tab with a single click. `PhishOracle` web app streamlines the process of generating phishing web pages for research and testing, offering a user-friendly interface for feature selection and web page generation.

7 Limitations

While our black-box approach demonstrates the effectiveness of visually transformed adversarial phishing web pages in evading detection models, it comes with some limitations. Our work focuses on a limited subset of transformation techniques and does not consider structural modifications such as background or layout changes. Although on the positive side, the selected transformations are simple to implement, visually deceptive to users, and less complex for real-world adversaries with limited knowledge. Additionally, we evaluated only one LLM-based phishing detection system. Evaluating against a broader set of LLMs is needed to better understand the generalizability of our approach to modern LLM-based phishing web page detections.

8 Conclusion

In this article, we developed `PhishOracle` to generate adversarial phishing web pages by embedding randomly selected content-based and visual-based phishing features into legitimate web pages. These adversarial phishing web pages were used to evaluate the robustness of the Stack model, VisualPhishNet, Phishpedia, and an LLM-based phishing detector. Our findings reveal that the Stack model exhibits reduced performance, while both VisualPhishNet and Phishpedia incorrectly identify the brands on adversarial phishing web pages due to various logo transformation techniques, leading to incorrect classifications. On the other hand, LLM-based phishing detection using Gemini (*LLM-PD*^{S}) demonstrates better performance against these adversarial attacks, although its detection capability also drops due to `PhishOracle`-generated phishing pages. These results highlight the vulnerabilities of existing phishing web page detection models, emphasizing the potential of MLLMs in advancing

phishing web page detection. Furthermore, our evaluation of tools from different security vendors on VirusTotal shows that adversarial phishing web pages generated by PhishOracle remain undetected within the first 24 hours, leaving them accessible to potential victims for an extended period. Moreover, the user study demonstrates that on an average ~48% of the PhishOracle-generated adversarial phishing web pages are misclassified as legitimate, by the participants. The datasets, PhishOracle code, and web app are available on our GitHub repositories [22, 23]. PhishOracle web app empowers users to clone and host it locally, enabling them to input a legitimate URL, choose specific phishing features, and generate corresponding phishing web pages.

The visual-based features in our tool add opacity to the logos and utilize transformation techniques such as rotation, blurring, adding gray-colored mesh, and noise on logos within a web page. In our future work, we plan to add a few more techniques to generate adversarial logos and incorporate them as a feature in PhishOracle tool.

Ethical Statement. Our institution does not require any formal IRB approval to carry out the research discussed here. Adhering to the principles outlined in the Menlo report [52], our user study is conducted with strict adherence to ethical guidelines, ensuring the non-forging and intentional avoidance of collecting sensitive information from participants. The study involved participants selecting an option ("Phishing" or "Legitimate") based on the visual appearance of web page screenshots. Although we plan to release the source code of PhishOracle tool on GitHub repositories [22, 23], rather than making it publicly available without restrictions, we will provide access on "Request Access" basis to ensure responsible use. It is intended solely for educational and research purposes and not for any illegal or unethical activities.

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